The pooled cloglog model, step by step

From one coin flip to a person's risk of developing coronary plaque

Learning notes for the CGPS (Herlev–Østerbro) who analysis

3 November 2026

How to read this

Every step adds one new piece of maths. Each piece comes with a picture and a small example with real numbers. All numbers are made up: three toy people, **A**, **B** and **C**, each with the nine trunk segments (proximal, mid and distal LAD, LCx and RCA), and one simulated toy cohort of 600 people. No CGPS data appear anywhere. The numbers are computed by `figures.py` with the same formulas as `CGPS_data/progression_who_pooled.py`, so the text, the figures and the code agree.
Section 17 puts the whole model on one page. Come back to it whenever you lose track. The covariates follow the *planned* design (SCORE2, segment position, log length, curvature, taper, the LM angle and the two LM diameter ratios). The box in Section 17 lists where the current code differs.

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1 Probability and a yes/no outcome

Idea

A **probability** is a number between 0 and 1 that says how often something happens in the long run. $p = 0.25$ means: among many people exactly like this one, about one in four.

For every person i and every segment j the outcome is a yes/no answer: did plaque appear in this segment between the baseline scan and the follow-up scan about ten years later? Write it as a number,

$$y_{ij} = \begin{cases} 1 & \text{plaque at follow-up (“yes”)} \\ 0 & \text{no plaque (“no”)} \end{cases}$$

The subscripts are labels: i says *which person*, j says *which segment*. $y_{\text{A,pLAD}}$ is the outcome of person A's proximal LAD.

A yes/no outcome with probability p of “yes” is called a **Bernoulli trial**: a coin that lands “yes” with probability p and “no” with probability $1 - p$. Each segment is its own coin, with its own p_{ij}.

Person A: chance that each segment develops plaque in 10 years

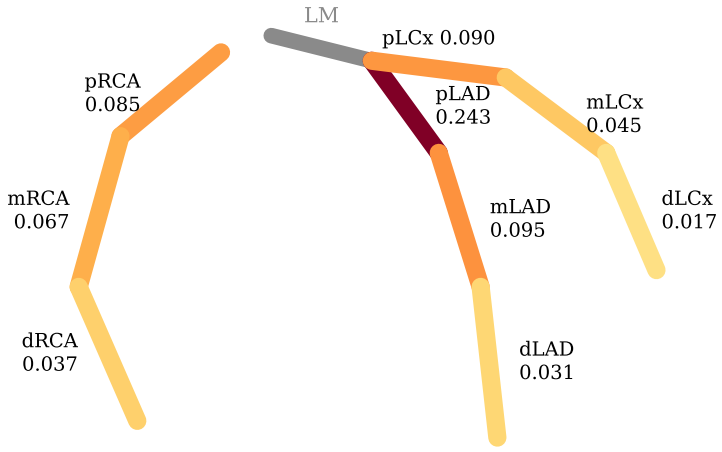


Figure 1. Person A's nine coins. Each number is the chance that the segment develops plaque within ten years (for now with the person effect b of Section 10 set to zero).

Worked example

Person A's pLAD has $p = 0.243$. Among 1,000 people exactly like A, about 243 would develop plaque in the pLAD. The chance of “no” is $1 - 0.243 = 0.757$.

One formula covers both answers, and it is used again for fitting in Section 15:

$$P(y) = p^{y}\,(1-p)^{1-y}.$$

For $y = 1$ it gives $p^1(1-p)^0 = p$; for $y = 0$ it gives $p^0(1-p)^1 = 1 - p$ (any number to the power 0 is 1).

Recap

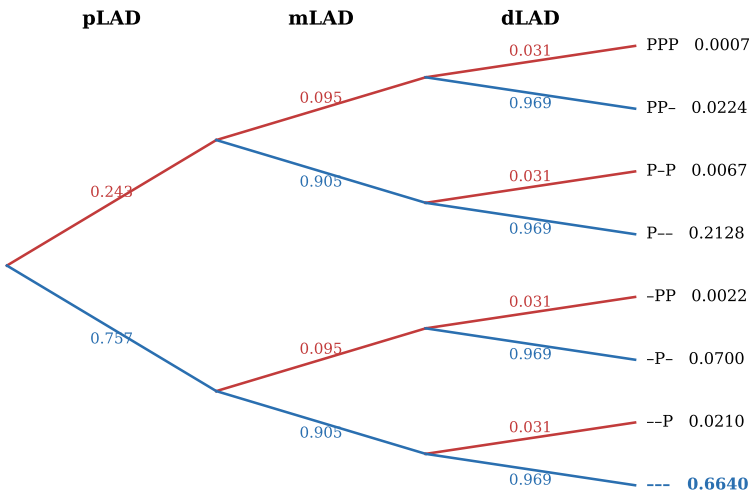
$y_{ij} \in \{0,1\}$; p_{ij} = chance of $y_{ij} = 1$; $P(y) = p^y(1-p)^{1-y}$.

2 Several segments, one person

The who question is about the *person*: does A develop plaque *anywhere* in the nine segments?

Idea

If two events do not influence each other (they are **independent**), the chance that both happen is the product of their chances. "Plaque nowhere" means "no" in segment 1 *and* "no" in segment 2 *and* ...



red = plaque (P), blue = no plaque (-). Only the path --- has no plaque anywhere: 0.6640. P(any plaque) = 1 − 0.6640 = 0.3360

Figure 2. All $2^3 = 8$ possible outcomes of A's three LAD segments. Multiply along a path to get its probability. Only one path has no plaque at all.

Worked example

For A's three LAD segments:

$$P(\text{no plaque in the LAD}) = (1-0.243)(1-0.095)(1-0.031) = 0.664,$$

so $P(\text{plaque somewhere in the LAD}) = 1 - 0.664 = 0.336$. It is easier to compute "none" and subtract from 1 than to add up the seven paths with at least one P.

The capital Greek letter Π ("pi") is shorthand for "multiply all of these":

$$P(\text{no plaque in any segment}) = \prod_{j=1}^{9}(1-p_{ij}), \qquad P(\text{any plaque}) = 1 - \prod_{j=1}^{9}(1-p_{ij}).$$

Careful

This product assumes the segments are independent. In real people they are not: someone whose pLAD develops plaque is more likely to develop it elsewhere too. Section 11 shows how the model repairs this, and Section 14 shows what goes wrong if you ignore it.

3 Rates, hazards and the exponential function

Plaque does not appear at the moment of the follow-up scan. It starts at some point during the ten years. The model describes this with a **hazard**: the rate at which plaque starts.

Idea

Chop the ten years into n short pieces. In each piece, plaque starts with a small chance. The total of these small chances over the window is the **cumulative hazard** Λ (capital "lambda"). With n pieces of chance Λ/n each, the chance that plaque never starts is $(1 - \Lambda/n)^n$.

Worked example

Take $\Lambda = 0.5$. The chance that plaque *does* start at some point, $1 - (1 - 0.5/n)^n$:

pieces n	1	2	10	100	1000	infinitely many
P(onset)	0.5000	0.4375	0.4013	0.3942	0.3935	0.3935

As the pieces get finer, the answer settles on $1 - \mathrm{e}^{-0.5} = 0.3935$.

This limit is where the number $\mathrm{e} \approx 2.718$ comes from. The rule is

$$\boxed{p = 1 - \mathrm{e}^{-\Lambda}}$$

e^x, also written $\exp(x)$, is the **exponential function**. Its partner is the **natural logarithm** log, which undoes it: $\log(\mathrm{e}^x) = x$ and $\mathrm{e}^{\log u} = u$.

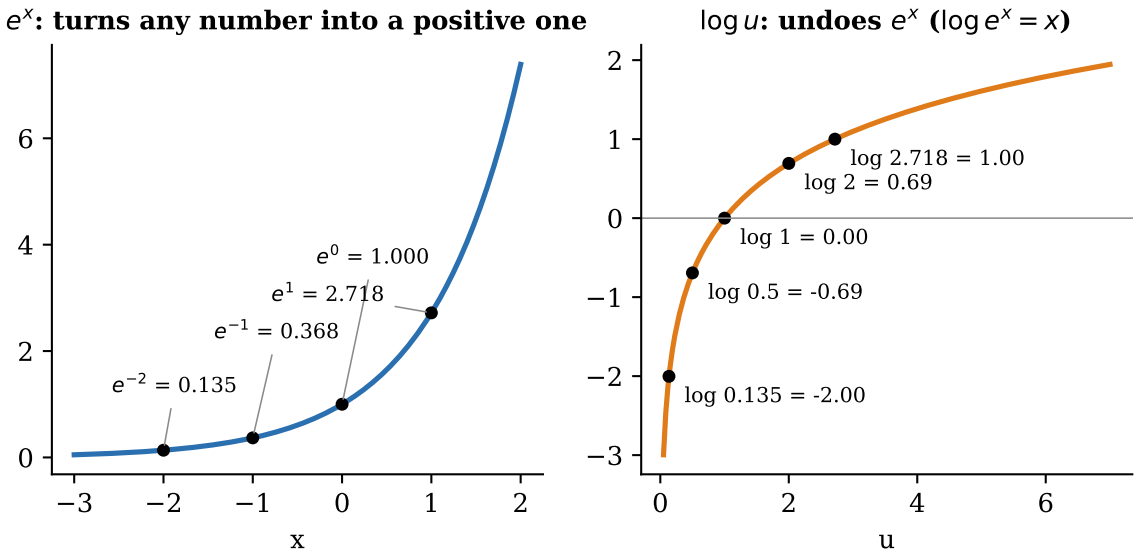


Figure 3. Left: e^x turns any number, also a negative one, into a positive one. Right: log goes back. $\log 1 = 0$, numbers below 1 have a negative log.

Three rules are used again and again below:

$$e^a\, e^b = e^{a+b}, \qquad \log(a \cdot b) = \log a + \log b, \qquad e^0 = 1.$$

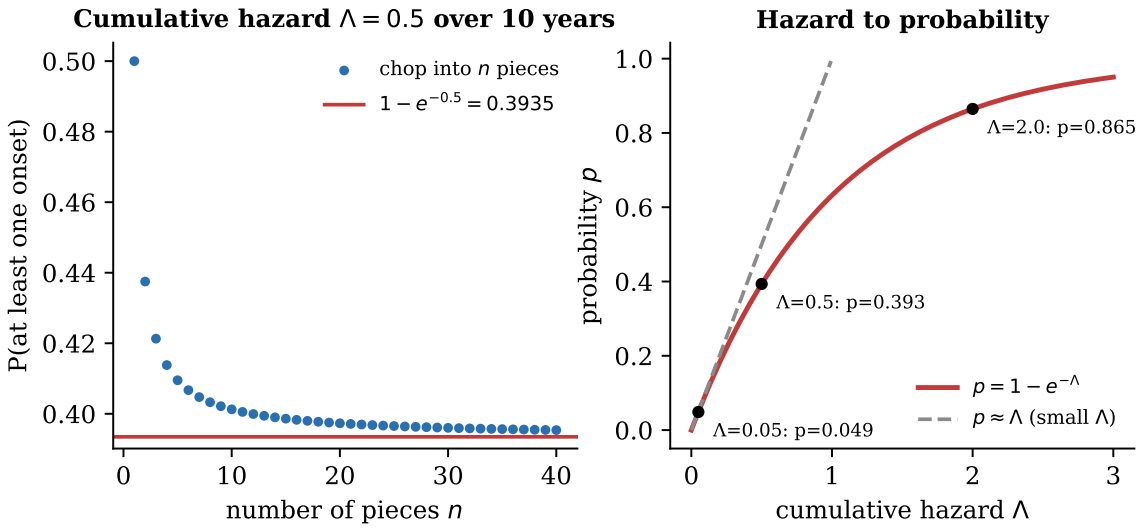


Figure 4. Left: chopping the window finer converges to $1 - e^{-\Lambda}$. Right: the curve $p = 1 - e^{-\Lambda}$. For small Λ, $p \approx \Lambda$; for large Λ, p flattens below 1.

Why the model needs this

A probability is stuck between 0 and 1, a hazard is not: it can be any positive number. Hazards can be multiplied by a risk factor without ever producing an impossible probability. One way to picture Λ: if plaque could start more than once, Λ is the expected number of starts in the window, and p is the chance of at least one (a Poisson process; [1]).

4 Hazards add up

Idea

The chance of no plaque in two segments is $e^{-\Lambda_1} \cdot e^{-\Lambda_2} = e^{-(\Lambda_1+\Lambda_2)}$. So two segments together behave like one segment whose hazard is the **sum** of the two.

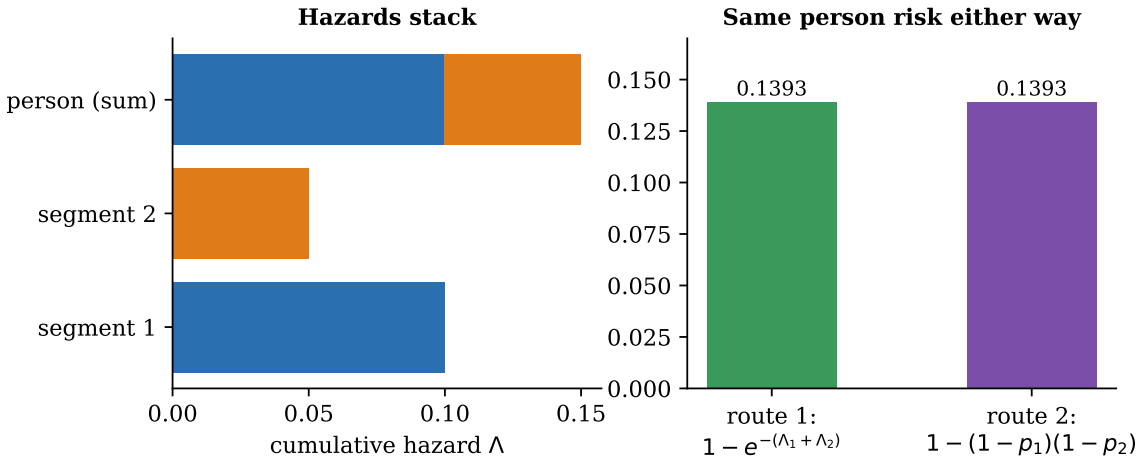


Figure 5. Adding hazards and multiplying "no" probabilities give the same person risk.

Worked example

Segment 1: $\Lambda_1 = 0.10$, so $p_1 = 1 - e^{-0.10} = 0.0952$. Segment 2: $\Lambda_2 = 0.05$, so $p_2 = 0.0488$.

- Route 1, add hazards: $1 - e^{-(0.10+0.05)} = 0.1393$.
- Route 2, multiply "no": $1 - (1 - 0.0952)(1 - 0.0488) = 0.1393$.

The capital Greek letter Σ ("sigma") means "add all of these". For person i, the **sum of the segment hazards** gets its own name:

$$S_i = \sum_{j=1}^{9} \Lambda_{ij}, \qquad P(\text{any plaque}) = 1 - e^{-S_i}.$$

Why the model needs this

S_i is the bridge from segments to the person. Section 12 shows that the person risk depends on the nine segments only through this one number.

5 Proportional hazards and the hazard ratio

Idea

A risk factor **multiplies** the hazard. If a segment has hazard ratio (HR) 1.5, its hazard is 1.5 times the hazard of an otherwise identical reference segment: $\Lambda = \Lambda_0 \times \text{HR}$. The ratio is the same whatever Λ_0 is. This is the **proportional hazards** assumption.

Worked example

HR $= 1.5$ at three baseline risks. "RR" is the risk ratio $p_{\text{new}}/p_{\text{ref}}$; the odds ratio (OR) compares odds $p/(1-p)$.

p_{ref}	$\Lambda_0 = -\log(1-p_{\text{ref}})$	$p_{\text{new}} = 1 - e^{-1.5\Lambda_0}$	RR	OR
0.01	0.0101	0.0150	1.50	1.50
0.10	0.1054	0.1462	1.46	1.54
0.50	0.6931	0.6464	1.29	1.83

The hazard ratio is 1.5 in every row. The risk ratio and the odds ratio drift with the baseline risk.

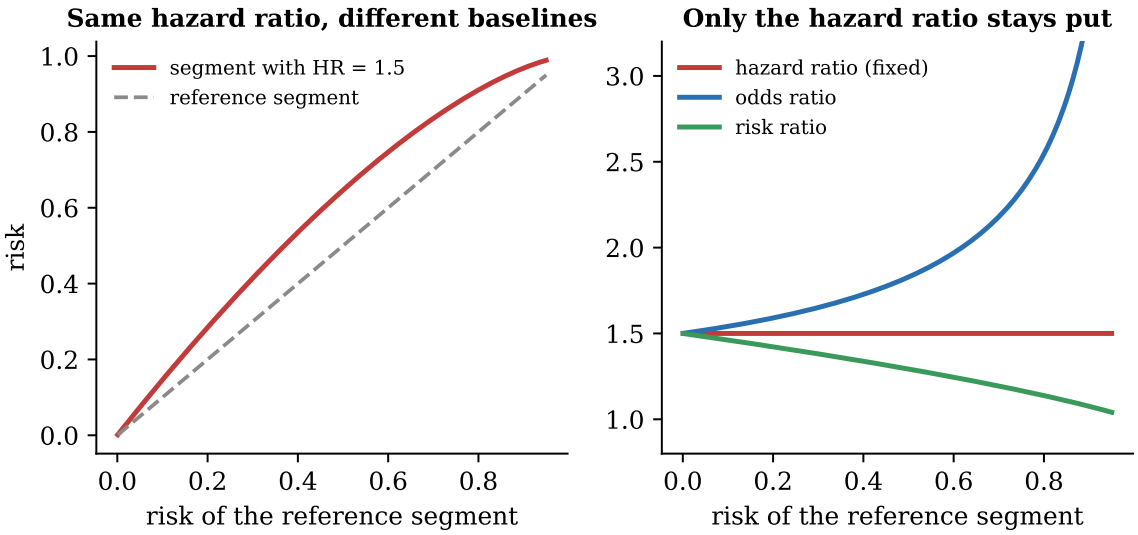


Figure 6. One and the same risk factor (HR $= 1.5$) reads as different odds ratios and risk ratios depending on how common plaque already is.

Why the model needs this

The nine segments have very different baseline risks (the pLAD is far more prone than the dLCx). A hazard ratio lets one coefficient mean the same thing in all nine. This model, for an event known only to have happened within an interval, is the grouped-data proportional hazards model of Prentice and Gloeckler [2].

6 The cloglog link: from a line to a probability

The model has to turn a sum of terms (any number, negative or positive) into a probability. It does so in two steps.

Idea

1. Call the sum of terms η ("eta"), the **linear predictor**. Make the hazard $\Lambda = e^{\eta}$, which is always positive.
2. Make the probability $p = 1 - e^{-\Lambda}$ (Section 3).

Together: $p = 1 - \exp(-e^{\eta})$. Undoing both steps gives $\eta = \log\{-\log(1-p)\}$, the **complementary log-log**, or **cloglog**, of p.

Worked example

η	$\Lambda = e^{\eta}$	cloglog $p = 1 - e^{-\Lambda}$	logit $p = 1/(1 + e^{-\eta})$
-3.0	0.0498	0.0486	0.0474
-1.0	0.3679	0.3078	0.2689
0.5	1.6487	0.8077	0.6225

At low risk ($\eta = -3$) cloglog and logit hardly differ. They separate at higher risk.

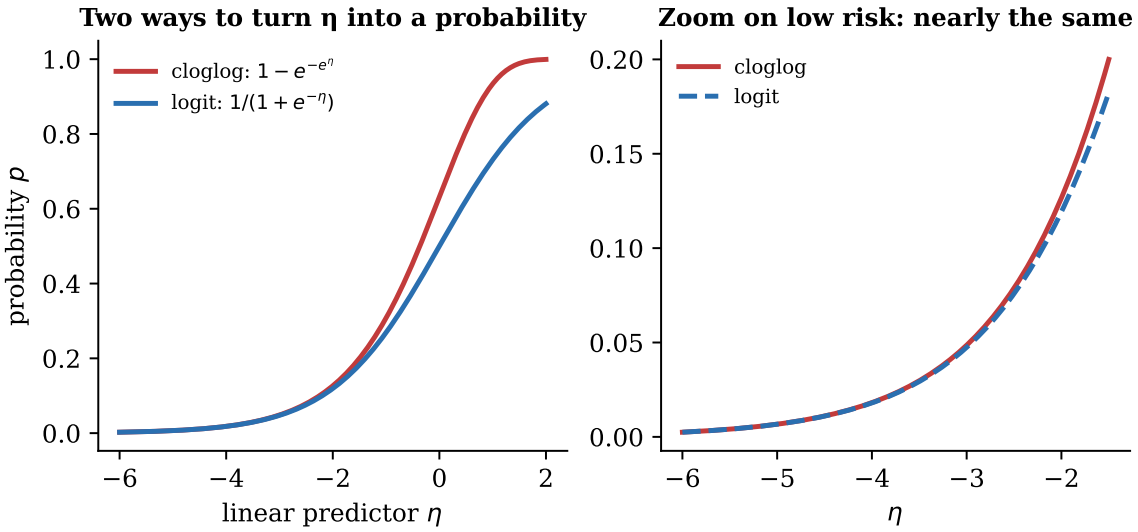


Figure 7. The cloglog curve and the more familiar logit curve. At low risk they nearly coincide.

Why the model needs this

The log turns "multiply the hazard" into "add to η": $\log(\Lambda_0 \cdot \mathrm{HR}) = \log \Lambda_0 + \log \mathrm{HR}$. So a coefficient β added to η multiplies the hazard by e^{β}: e^{β} **is a hazard ratio**. With the logit link, e^{β} would be an odds ratio, which drifts with the baseline risk (Figure 6). The cloglog link is the usual choice for rare binary outcomes and for this hazard reading [1, 2].

7 The linear predictor: adding up the terms

Idea

η_{ij} is a weighted sum. Each covariate value x is multiplied by its coefficient β ("beta"), and the results are added to a starting value α_j ("alpha") for the segment type:

$$\eta_{ij} = \alpha_j + \beta_1 x_{ij,1} + \beta_2 x_{ij,2} + \dots$$

The terms in the planned model:

- α_j: one value per segment type (nine in total). It is the log hazard of that segment for a reference person with all covariates at 0. It carries "plaque is more common proximally and in the LAD".
- SCORE2 (z-scored, Section 8): the same value on all nine rows of a person.
- log length: longer segments have more wall where plaque can start. If its coefficient were exactly 1, hazard would be proportional to length; the model estimates it freely.
- curvature and taper (z within segment type): a different value on every row.
- the LM angle and the LM→LAD and LM→LCx diameter ratios: one value per person, entered only on the rows they can act on (Section 9).

In the toy model the coefficients are $\beta_{\text{SCORE2}} = 0.50$, $\beta_{\text{length}} = 0.60$, $\beta_{\text{curv}} = 0.20$, $\beta_{\text{taper}} = 0.15$, $\beta_{\text{LM angle}} = 0.25$ and $\beta_{\text{LM}\to\text{LAD}} = \beta_{\text{LM}\to\text{LCx}} = 0.10$.

Worked example: Person A's pLAD

term	value x	coefficient β	$\beta \cdot x$
α_{pLAD}			-2.2
SCORE2 (z)	+0.5	0.50	+0.250
log length	+0.1	0.60	+0.060
curvature (z)	+1.2	0.20	+0.240
taper (z)	+0.6	0.15	+0.090
LM angle (z)	+1.0	0.25	+0.250
LM→LAD ratio (z)	+0.3	0.10	+0.030
LM→LCx ratio (z)	off on LAD rows	0.10	0
$\eta_{\text{A,pLAD}}$			-1.28

Hazard: $\Lambda = \mathrm{e}^{-1.28} = 0.2780$. Probability: $p = 1 - \mathrm{e}^{-0.2780} = 0.243$.

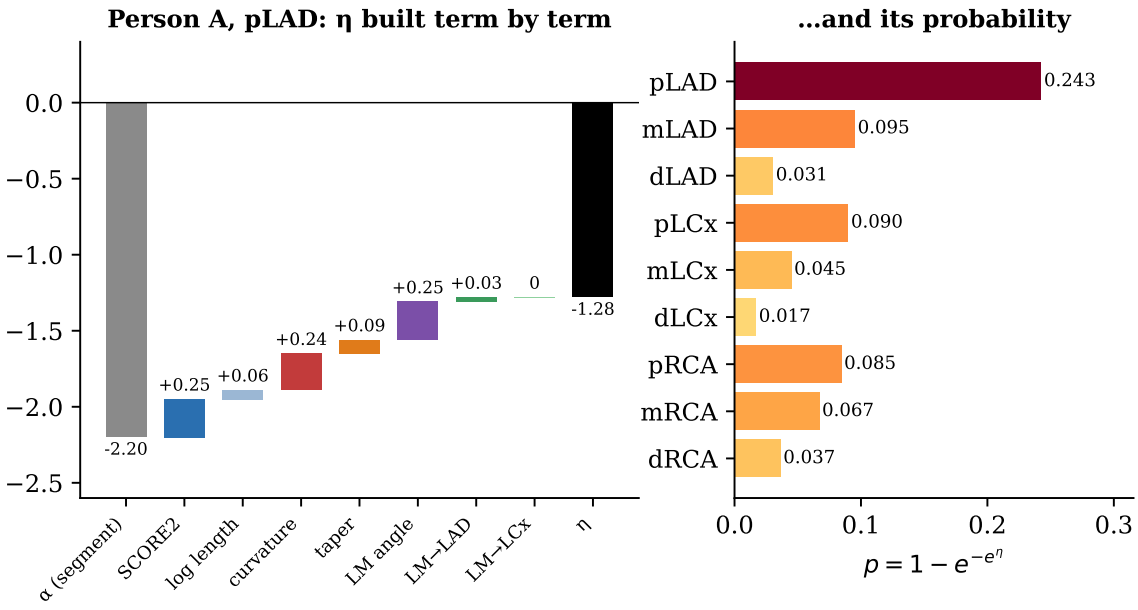


Figure 8. Left: the same calculation as a waterfall. Each bar moves η up or down from α_{pLAD}. Right: the probabilities of all nine of A's segments.

Recap

Compact notation: stack the covariates in a list (a **vector**) $\mathbf{x}_{ij}$ and the coefficients in $\boldsymbol{\beta}$. Then $\eta_{ij} = \alpha_j + \mathbf{x}_{ij}^{\top}\boldsymbol{\beta}$, where $\mathbf{x}^{\top}\boldsymbol{\beta}$ means "multiply pairwise and add". Reading a coefficient: one unit more of x multiplies the segment hazard by e^{β}. For SCORE2 that is $e^{0.50} = 1.649$ per SD.

8 z-scores within segment type

Curvature means different things in different segments. Distal segments bend more than proximal ones, so "0.11 per mm" is a lot in the pLAD and ordinary in the dRCA.

Idea

A **z-score** says how many standard deviations (SD) a value lies above or below the average:

$$z_{ij} = \frac{x_{ij} - m_j}{s_j},$$

where m_j and s_j are the mean and SD of x *in segment type* j. The SD measures the typical spread around the mean.

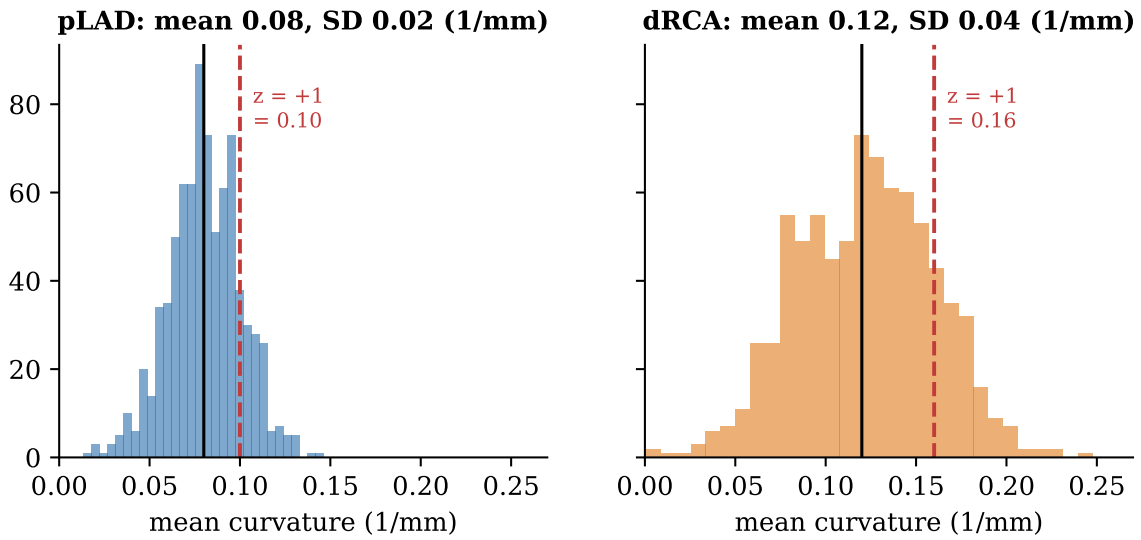


Figure 9. Made-up curvature distributions for the pLAD and the dRCA. The red line marks $z = +1$ in each.

Worked example

A curvature of 0.11 per mm:

- in a pLAD (mean 0.08, SD 0.02): $z = (0.11 - 0.08)/0.02 = 1.5$, a strongly bent pLAD;
- in a dRCA (mean 0.12, SD 0.04): $z = (0.11 - 0.12)/0.04 = -0.25$, a slightly straighter than usual dRCA.

Why the model needs this

One shared coefficient on z means: "1 SD more curved than usual *for this segment type*" carries the same hazard ratio everywhere. In raw units the effect differs by segment: the hazard ratio per 0.01 per mm is $\exp(\beta \cdot 0.01/s_j)$, that is $\exp(0.10) = 1.105$ in the pLAD and $\exp(0.05) = 1.051$ in the dRCA. Segment types with a narrow spread get a steeper raw slope. The mean m_j is absorbed by α_j anyway; the scaling by s_j is the real choice.

Careful

m_j and s_j must come from the people the model is trained on. In cross-validation they are recomputed in every training fold.

9 The geometry terms

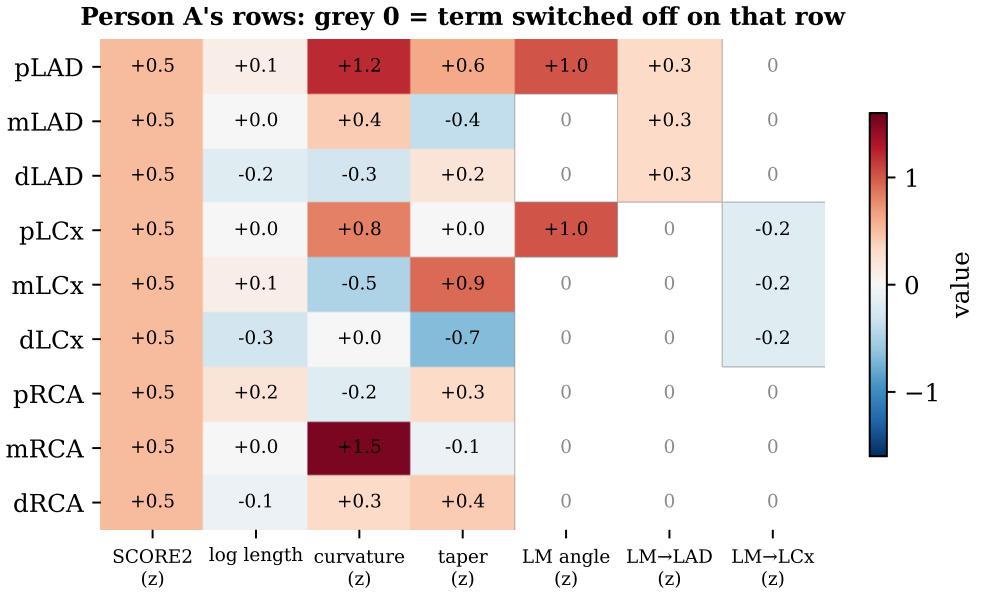


Figure 10. Person A's nine rows. Every row gets SCORE2, log length, curvature and taper. The LM angle sits only on the pLAD and pLCx rows, the LM→LAD ratio only on LAD rows, the LM→LCx ratio only on LCx rows. A grey 0 switches the term off on that row.

Idea

Curvature and taper are measured per segment, so they enter every row. The LM bifurcation features are measured once per person and enter only on the rows downstream of the bifurcation they describe. Putting 0 on the other rows fixes their coefficient there at zero, which costs no extra parameter.

Worked example

The per-SD hazard ratios of the toy model: curvature $e^{0.20} = 1.221$, taper $e^{0.15} = 1.162$, LM angle $e^{0.25} = 1.284$, each diameter ratio $e^{0.10} = 1.105$. A pLAD that is 1 SD more curved than usual has 1.221 times the hazard of an otherwise identical pLAD of the same person.

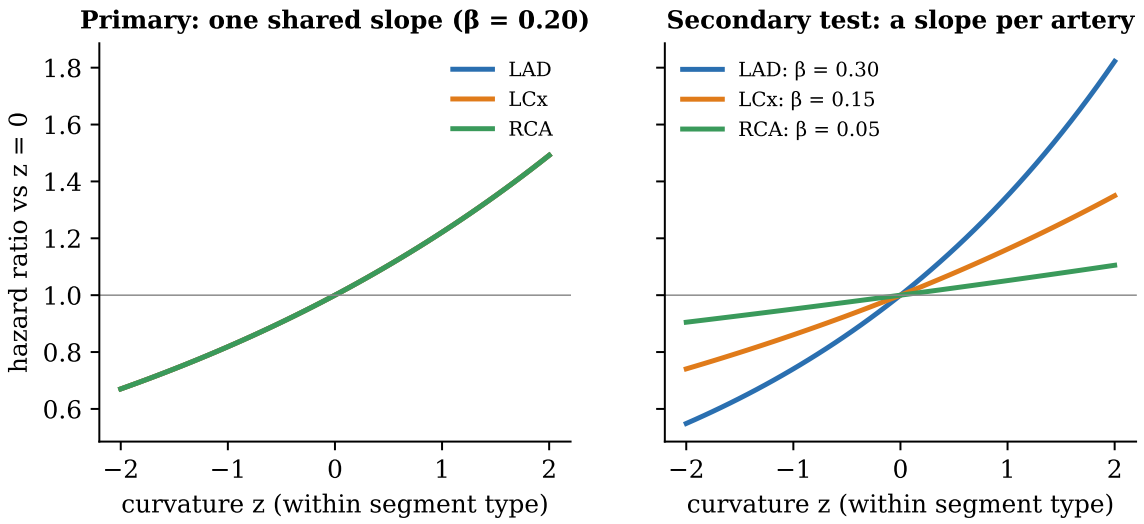


Figure 11. Left: the primary model, one slope for all arteries. Right: the planned secondary test, one slope per artery for curvature and taper (two extra slopes each, 4 extra parameters).

Why the model needs this

The five geometry terms together form the **geometry block** (5 parameters). The one prespecified test of the who question asks whether this block improves the model (Section 16). Whether geometry acts differently by artery is a separate, secondary test, so it cannot change the primary answer.

10 People differ: the random intercept

Two people with the same SCORE2 and the same geometry can still differ in risk: genes, diet and many things nobody measured. The model gives every person an extra term b_i:

$$\eta_{ij} = \alpha_j + \mathbf{x}_{ij}^{\top}\boldsymbol{\beta} + b_i.$$

Idea

b_i is unknown, so the model does not estimate it person by person. It assumes the b_i are spread like a **normal distribution** (the bell curve) with mean 0 and SD σ, written $b_i \sim N(0, \sigma^2)$. Only σ is estimated. About 68% of people lie within $\pm\sigma$ of 0, about 95% within $\pm 2\sigma$.

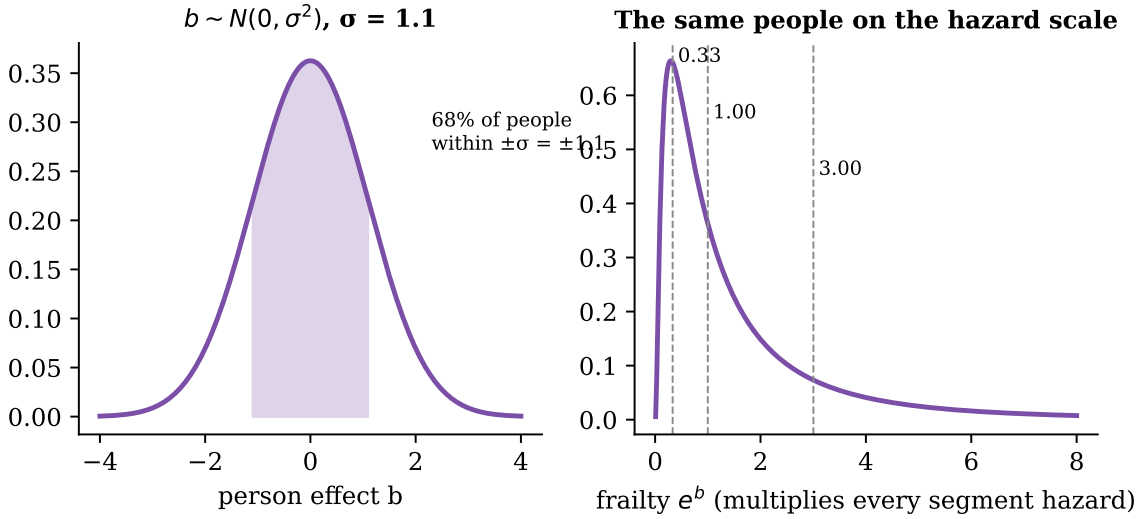


Figure 12. Left: the spread of person effects in the toy model, $\sigma = 1.1$. Right: the same spread as a multiplier e^b on the hazard, called the **frailty**.

Since $e^{\eta+b} = e^{\eta} \cdot e^{b}$, the person effect multiplies *all nine* hazards of a person by the same factor e^{b_i}. In survival analysis this is a **shared frailty** [3].

Worked example

$b = +1$ multiplies every hazard by $e^1 = 2.718$; $b = -1$ by $e^{-1} = 0.368$. For A's pLAD: $p = 0.097$ at $b = -1$, 0.243 at $b = 0$, 0.530 at $b = +1$. With $\sigma = 1.1$, the middle 68% of people have frailties between $e^{-1.1} = 0.33$ and $e^{1.1} = 3.00$.

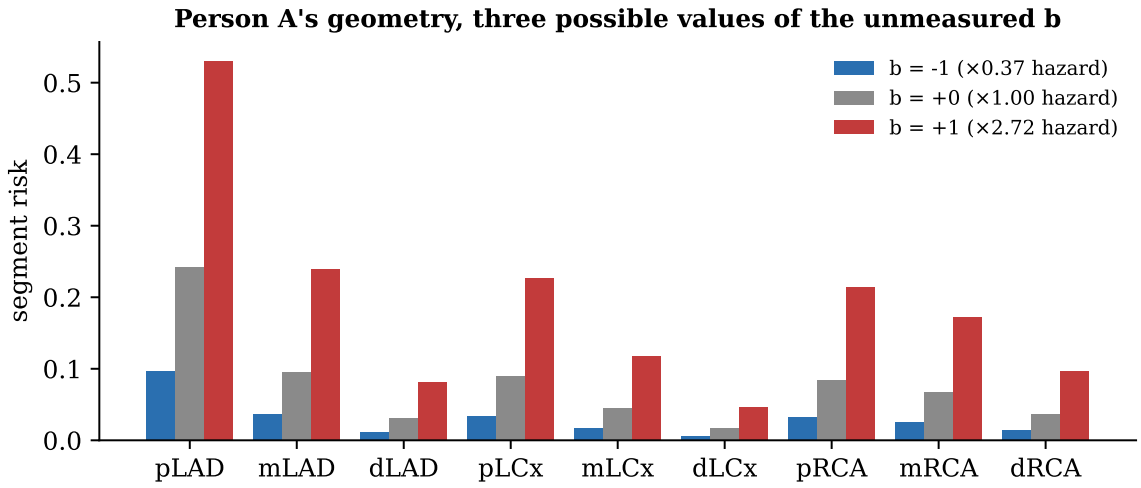


Figure 13. The same geometry, three values of the unmeasured person effect.

Why the model needs this

σ is hard to picture. The **median hazard ratio** translates it: pick two random people with identical covariates and divide the higher hazard by the lower; the median of that ratio is $\text{MHR} = \exp\left(\sqrt{2\sigma^2} \times 0.6745\right) = 2.86$ for $\sigma = 1.1$ [4]. (0.6745 is the point that cuts off the upper quarter of a standard bell curve.)

11 Why segments of one person go together

Idea

Given b_i, the nine coins of person i are independent. But b_i is shared. If one segment of a person developed plaque, that hints that b_i is high, which raises the chance for the others. Without knowing b_i, the segments are **positively correlated**.

Worked example

Two segments with $\eta = -2.5$ each and $\sigma = 1$ (simulated with 400,000 people): $P(\text{both}) = 0.0266$, while $P(\text{seg 1}) \times P(\text{seg 2}) = 0.0135$. Both-segment plaque is about 2.0 times as common as independence predicts.

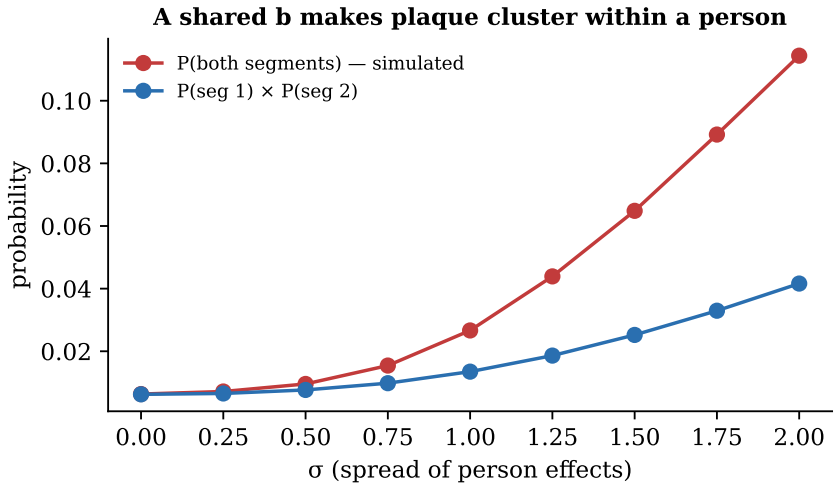


Figure 14. The larger σ, the more plaque clusters within people. At $\sigma = 0$ the two lines meet.

Why the model needs this

Treating the 5,400 rows of a cohort as independent would overstate how much information the data hold. One alternative, generalized estimating equations (GEE), corrects the standard errors but gives no person risk formula [5]. The random intercept models the dependence directly, and that is what makes the next step possible.

12 From segments to a person, for a known b

Fix b for a moment. Then the segments are independent, and Sections 2–4 apply:

$$P(\text{no plaque anywhere} \mid b) = \prod_j \exp\left(-e^{\eta_{ij}+b}\right) = \exp\left(-e^b \sum_j e^{\eta_{ij}}\right) = \exp\left(-e^b S_i\right).$$

The vertical bar "$\mid b$" reads "given b". $S_i = \sum_j e^{\eta_{ij}}$ is the sum of the nine segment hazards at $b = 0$.

Idea

For a known b, "any plaque in the person" is itself a cloglog model, with hazard $e^b S_i$. The nine segments matter only through their summed hazard.

Worked example

Person A: $S_A = 0.7627$. The pLAD alone contributes 0.2780, about 36% of it.

- $b = -1$: $1 - \exp(-0.368 \times 0.7627) = 0.245$
- $b = 0$: $1 - \exp(-0.7627) = 0.534$
- $b = +1$: $1 - \exp(-2.718 \times 0.7627) = 0.874$

Why the model needs this

The share explains why local geometry moves person risk only a little. If the pLAD had a segment HR of 1.3, A's summed hazard would grow by a factor $(1 - 0.36) + 0.36 \times 1.3 \approx 1.11$, not 1.3. Person-level gains from segment geometry are diluted by the segments it does not touch.

13 Averaging over the unknown b: an integral

For a new person, b is unknown. The person risk is the average of $P(\text{any} \mid b)$ over all possible b, each weighted by how common it is:

$$\boxed{P_i = 1 - \int \exp\left(-e^b S_i\right) \phi(b; 0, \sigma^2)\, db}$$

$\phi(b; 0, \sigma^2)$ is the height of the bell curve at b.

Idea

An **integral** $\int f(b)\, db$ is the area under a curve. Picture it as many thin bars of width db and height $f(b)$, added up. Here it is a weighted average: likely values of b count a lot, rare ones a little.

The integral has no formula you can write down, so it is computed numerically. **Gauss–Hermite quadrature** replaces the curve by a handful of well-chosen points b_k with weights w_k [6]:

$$P_i \approx \sum_k \tilde{w}_k\, P(\text{any} \mid b_k), \qquad b_k = \sqrt{2}\,\sigma\, x_k, \quad \tilde{w}_k = w_k / \sqrt{\pi},$$

where x_k and w_k are fixed numbers from a table. The $\tilde{w}_k$ add up to 1.

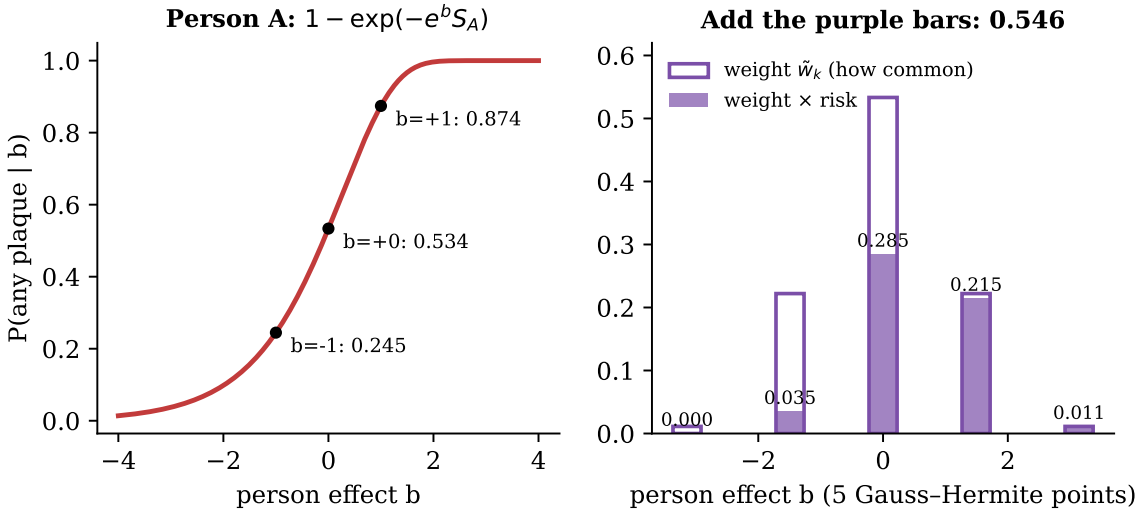


Figure 15. Left: A's risk for every possible b. Right: the five Gauss–Hermite points. Each weight (outline) times the risk at that point (filled) gives one purple bar; their sum is the person risk.

Worked example: Person A with 5 points ($\sigma = 1.1$)

k	x_k	$b_k = \sqrt{2}\sigma x_k$	$\tilde{w}_k$	$P(\text{any} \mid b_k)$	$\tilde{w}_k \times P$
1	-2.020	-3.143	0.0113	0.0324	0.0004
2	-0.959	-1.491	0.2221	0.1578	0.0350
3	+0.000	+0.000	0.5333	0.5336	0.2846
4	+0.959	+1.491	0.2221	0.9662	0.2146
5	+2.020	+3.143	0.0113	1.0000	0.0113
			1.0000	sum	0.5458

With 40 points: $P_A = 0.5476$; with 200 points: 0.5476. Five points already land close; the code uses 40.

Recap

Person risk = average over the unknown person effect of the risk given that effect. It is exact under the model [3]; the only approximation is the number of quadrature points, which is checked.

14 Two tempting shortcuts, and why both fail

Idea

Shortcut 1: set $b = 0$. Zero is the *median* person effect, so this gives the median person's risk, not the average risk [7].
Shortcut 2: multiply marginal segment risks. First average each segment over b separately, then combine them as if independent: $1 - \prod_j (1 - \bar{p}_{ij})$. This forgets that the segments share b (Section 11).

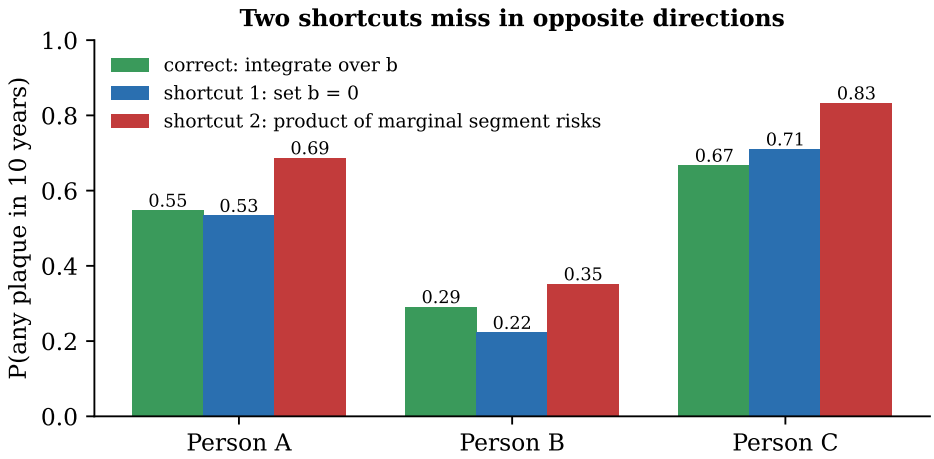


Figure 16. The correct person risk (green) and the two shortcuts for the three toy people.

Worked example

	S_i	correct P_i	$b = 0$	naive product
Person A	0.7627	0.548	0.534	0.687
Person B	0.2516	0.291	0.222	0.352
Person C	1.2387	0.667	0.710	0.833

A's marginal pLAD risk is $\bar{p} = 0.311$, higher than 0.243 at $b = 0$: averaging over b lifts every segment, and the product then counts the shared part of that lift nine times.

Why the model needs this

The naive product is *always* too high, for every person. Setting $b = 0$ is too low for low-risk people (A, B) and too high for high-risk ones (C); roughly, the switch happens near 0.63. Discrimination (who is ranked above whom) does not detect the $b = 0$ error, only calibration does. New people must be predicted by integrating over b [7, 8].

15 Fitting the model: the likelihood

So far the coefficients were given. In practice they are estimated from data.

Idea

The **likelihood** of a candidate value is the probability of the data you actually saw, if that value were true. The best estimate is the value with the highest likelihood (**maximum likelihood**).

Worked example: one coin

10 segments, 3 with plaque. For a candidate p, the probability of this result is $L(p) = p^3(1-p)^7$ (Section 1, multiplied over the 10 coins).

candidate p	0.1	0.3	0.5
$L(p)$	0.000478	0.002224	0.000977
$\log L(p)$	-7.65	-6.11	-6.93

$p = 0.3$ wins, which is just $3/10$.

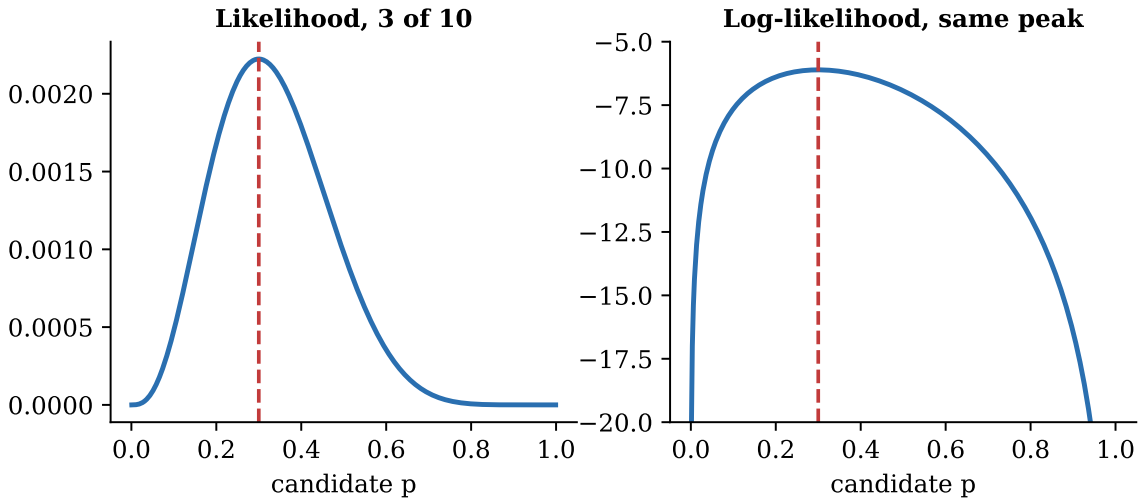


Figure 17. The likelihood and its log peak at the same place. Logs turn tiny products into manageable sums.

The full model. For one person with a known b, the probability of their nine outcomes is a product over segments. b is unknown, so it is integrated out, exactly as in Section 13:

$$L_i(\boldsymbol{\beta}, \sigma) = \int \prod_j p_{ij}(b)^{y_{ij}} \{1 - p_{ij}(b)\}^{1-y_{ij}} \, \phi(b; 0, \sigma^2) \, \mathrm{d}b, \qquad p_{ij}(b) = 1 - \exp(-\mathrm{e}^{\eta_{ij}+b}).$$

People are independent, so the log-likelihood of the cohort is the sum $\ell(\boldsymbol{\beta}, \sigma) = \sum_i \log L_i$. A computer searches for the $(\boldsymbol{\beta}, \sigma)$ with the largest ℓ, walking uphill step by step (the code uses L-BFGS-B with the integral done by Gauss–Hermite quadrature). Quadrature centred at the mode of each person's integrand (*adaptive*) is more accurate with few points [9]; the code uses fixed points and checks the result by doubling them.

Worked example: the toy cohort

600 simulated people, 5,400 segment rows, 492 segments with plaque, 267 people (44%) with plaque somewhere. Fitted against the true values:

	SCORE2	length	curv	taper	LM angle	LM→LAD	LM→LCx	σ
true	0.50	0.60	0.20	0.15	0.25	0.10	0.10	1.1
estimate	0.50	0.87	0.25	0.18	0.26	0.15	0.20	1.08

The estimates scatter around the truth. With a different random cohort they would scatter differently; that scatter is what standard errors and confidence intervals describe.

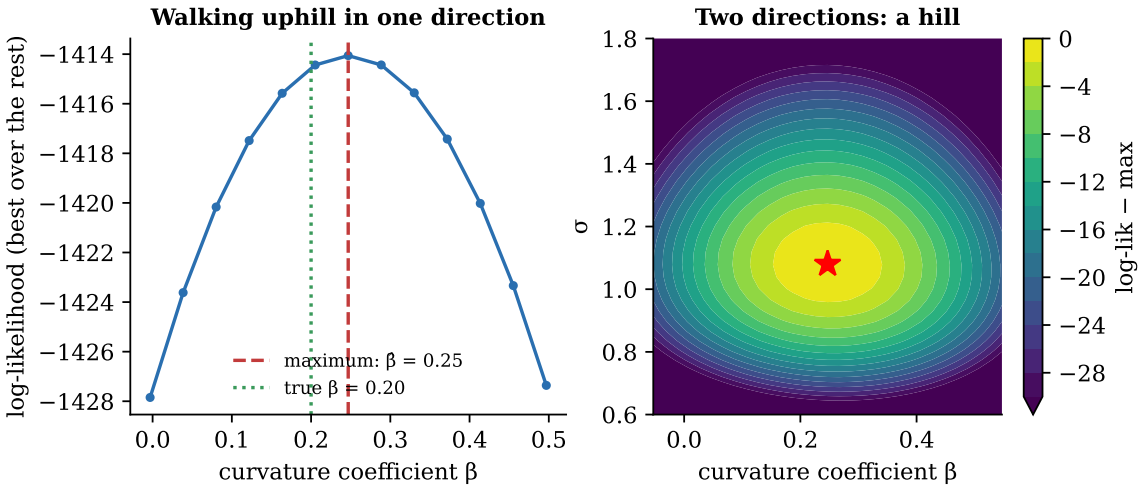


Figure 18. Left: the log-likelihood as the curvature coefficient is moved, with everything else re-optimized. Right: the hill over two directions at once (curvature coefficient and σ, the rest held at the peak). The star is the maximum.

16 Testing geometry: the likelihood-ratio test

Idea

Fit two models that differ only in the geometry block:

- **base**: segment intercepts, SCORE2, log length, σ;
- **full**: base plus the 5 geometry terms.

The full model always fits at least as well. The question is whether the gain is larger than chance would give. The **likelihood-ratio statistic** is $\mathrm{LR} = 2(\ell_{\mathrm{full}} - \ell_{\mathrm{base}})$. If geometry does nothing, LR follows a **chi-square (χ^2) distribution** with degrees of freedom (df) equal to the number of added terms.

Worked example

Toy cohort: $\ell_{\mathrm{full}} = -1414.06$, $\ell_{\mathrm{base}} = -1441.28$, so $\mathrm{LR} = 2 \times (-1414.06 - (-1441.28)) = 54.44$ on 5 df. If geometry did nothing, LR would exceed 11.07 only 5% of the time. The observed LR is far beyond that: $p = 1.7 \times 10^{-10}$ (the chance of an LR this large if geometry did nothing).

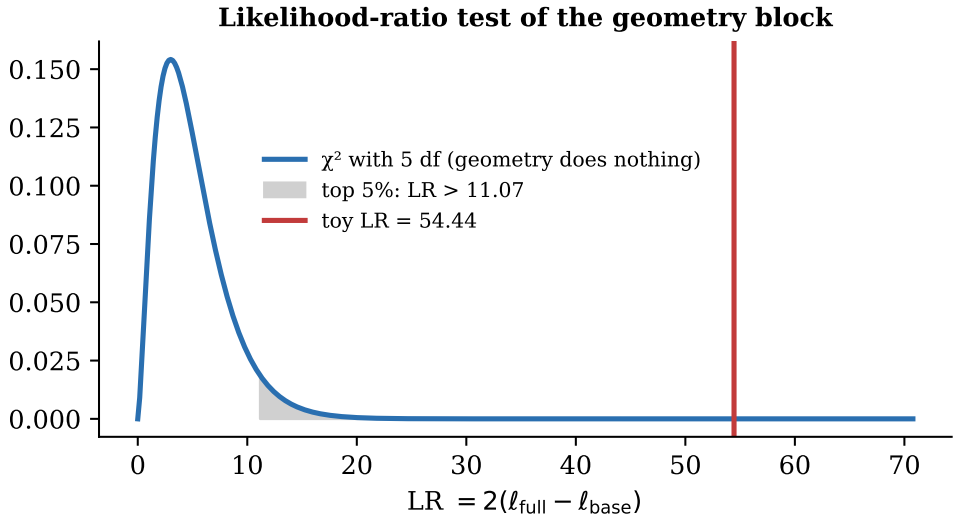
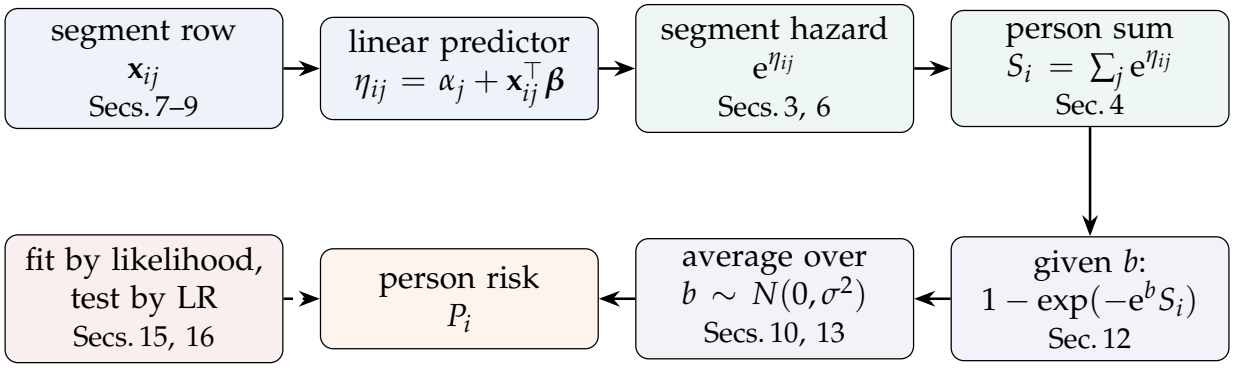


Figure 19. What LR looks like when geometry does nothing (blue), the top 5% (grey) and the toy result (red).

Careful

- The toy cohort was built with large geometry effects. In the real cohort the effect is expected to be small.
- Both models must be fitted on the same rows, with the same link and the same number of quadrature points.
- σ is in both models, so the plain χ^2 applies. Only a test of $\sigma = 0$ itself would need a different reference distribution [10].
- The test uses both who-information (between people) and where-information (which segments within a person). A rejection says geometry is linked to segment hazards; whether it helps rank *people* is judged by the person-level evaluation, not by this test.

17 The full model on one page



The model

For person i and trunk segment j (pLAD, mLAD, dLAD, pLCx, mLCx, dLCx, pRCA, mRCA, dRCA):

$$\begin{aligned}\eta_{ij} &= \alpha_j + \beta_{\mathrm{S}}\,\mathrm{SCORE2}_i + \beta_{\mathrm{L}}\log \mathrm{length}_{ij} + \beta_{\mathrm{c}}\, z_{ij}^{\mathrm{curv}} + \beta_{\mathrm{t}}\, z_{ij}^{\mathrm{taper}} \\ &\quad + \beta_{\mathrm{a}}\, z_i^{\text{LM angle}}\,[j \in \{\mathrm{pLAD}, \mathrm{pLCx}\}] \\ &\quad + \beta_{\mathrm{d}}\, z_i^{\mathrm{LM}\to\mathrm{LAD}}\,[j \in \mathrm{LAD}] + \beta_{\mathrm{x}}\, z_i^{\mathrm{LM}\to\mathrm{LCx}}\,[j \in \mathrm{LCx}] \\ P(y_{ij} = 1 \mid b_i) &= 1 - \exp\!\big(-\mathrm{e}^{\eta_{ij}+b_i}\big), \qquad b_i \sim N(0,\sigma^2) \\ P_i = P(\text{any plaque}) &= 1 - \int \exp\!\Big(-\mathrm{e}^{b}\textstyle\sum_j \mathrm{e}^{\eta_{ij}}\Big)\,\phi(b;0,\sigma^2)\,\mathrm{d}b\end{aligned}$$

$[\,\cdot\,]$ is 1 when the condition holds and 0 otherwise. e^{β} is a segment hazard ratio, for a given person, per unit (per within-type SD for the z terms). Geometry block: $\beta_{\mathrm{c}}, \beta_{\mathrm{t}}, \beta_{\mathrm{a}}, \beta_{\mathrm{d}}, \beta_{\mathrm{x}}$, 5 df.

What "where" can be read from this model

- The intercepts α_j: where plaque tends to arise, geometry aside.
- The geometry hazard ratios: within one person, a segment that is 1 SD more curved has $\mathrm{e}^{\beta_{\mathrm{c}}}$ times the hazard.
- Ranking segments within a person: b_i is the same on all of a person's rows, so it cancels from comparisons between them. The hazard ratio of A's pLAD versus A's dLCx is $\mathrm{e}^{\eta_{\mathrm{pLAD}}-\eta_{\mathrm{dLCx}}} = \mathrm{e}^{-1.28-(-4.05)}$, whatever b_{A} is. A within-person C statistic summarises this ranking.
- The secondary curvature/taper × artery test (Figure 11).

These are descriptive. Caveat: $\boldsymbol{\beta}$ is estimated from comparisons both between and within people, so it is not a purely within-person effect.

Where the current code differs (`progression_who_pooled.py`)

- Data and outcome: Medis trees (people with plaque at baseline); outcome is *progression* (higher grade at follow-up), not incident plaque in the plaque-free tracer cohort.
- Base (B2): log follow-up, log SCORE2 (z), dominance, segment position, log length and baseline grade.
- Geometry (G02): mean turning angle T5 (z within segment type) and the LM angle on the pLAD and pLCx rows: 2 df. No taper and no LM diameter ratios.
- Numerics: 30 Gauss–Hermite points for fitting by default, 40 for the person risk.

18 What the model assumes

1. **Proportional hazards.** A covariate multiplies the hazard by the same factor in every segment and over the whole window (Section 5). Per-artery slopes (Figure 11) are one check.
2. **Normal person effects.** $b_i \sim N(0,\sigma^2)$. A gamma frailty is the usual alternative; it has a

closed form $P_i = 1 - (1 + \theta S_i)^{-k}$ (shape k, scale θ) [3]. With the same mean and variance of the frailty as the toy model ($k = 0.425$, $\theta = 4.310$), A's risk would be 0.461 instead of 0.548. Refitted to data, σ and $\boldsymbol{\beta}$ would adjust and the gap would shrink; this is a sensitivity analysis, not an estimate of the error.

3. **One shared person effect.** Segments are independent given b_i. A second level, such as an extra effect per artery within a person, is not in the model.
4. **Straight lines in η.** Each extra SD of curvature multiplies the hazard by the same factor. A U-shape would be missed; a spline check can test it.
5. **Person effect unrelated to the covariates.** b_i is assumed independent of geometry. If tortuous vessels go with unmeasured risk, part of β reflects who rather than where.

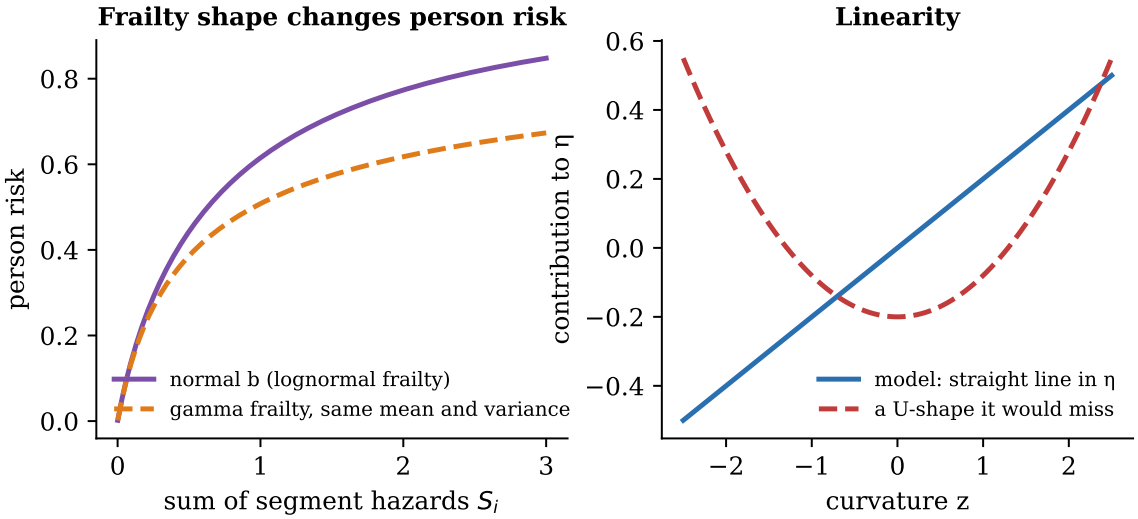


Figure 20. Left: the frailty shape changes the person risk for the same S_i. Right: a U-shaped effect that a straight line in η cannot follow.

A Symbols

symbol	meaning
i, j	person, segment
y_{ij}	1 if segment j of person i has plaque at follow-up, else 0
p_{ij}	probability that $y_{ij} = 1$
Λ	cumulative hazard over the window; $p = 1 - \mathrm{e}^{-\Lambda}$
η_{ij}	linear predictor (log hazard); $\Lambda = \mathrm{e}^{\eta}$
α_j	segment-type intercept
$\boldsymbol{\beta}$, β_k	coefficients; e^{β_k} = hazard ratio per unit
z	z-score, (value − mean)/SD
b_i, σ	person effect and its SD; e^{b_i} = frailty
S_i	sum of the segment hazards of person i
P_i	person risk of any plaque
$\phi(b; 0, \sigma^2)$	bell-curve height at b
$\sum$, $\prod$, $\int$	add up, multiply, area under a curve
ℓ	log-likelihood; LR $= 2(\ell_{\text{full}} - \ell_{\text{base}})$

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